

Passages - Advanced Portfolio Management

Oftentimes, we assume that returns are normally distributed. This is an optimistic assumption, but it provides a useful reference point, in the sense that changes in volatility estimations roughly correspond to changes in losses that could be experienced with a given probability.

A loss of minus one standard deviation or more from the expected return occurs with a probability of about 16%. In one year (or 252 trading days), you can expect 40 days with such returns.

Valuation Analysis: This differentiated view comes from a company business model whose data are coming primarily from Cash Flow Statements, Balance Sheets (Statements of Financial Positions), Income Statements and Stakeholder Equity Statements. It is complemented by macroeconomic data affecting the production function of the company as well as demand for its products and services. This is the domain of fundamental analysis [T. Koller and Wessels, 2015] and affects both short-term forecasts, e.g., earnings or the event of a sell-side analyst downgrading or upgrading a recommendation, and long-term forecast, such as the potential long-term unsuitability of a business.

The variance of the sum of two independent random variables is equal to the sum to the variances of the two random variables. You can interpret the standard deviations as sides of a right triangle (i.e. Pythagoras' Theorem)

The alpha term is a constant, while the

terms βx_m and Error are random. The constant α does not contribute to volatility, but the two random terms do.

The inflation-adjusted annualized historical return of the S&P 500 from 1926 (the inception year of the index) to 2018 is 7%; so it may seem a good idea to have a positive beta to the market.

One less attractive feature of carrying beta in your portfolio is that it is harder and harder to justify to investors. The principals who entrusted their capital to you also see the alpha-beta decomposition; and they usually have very inexpensive ways to invest in the market (i.e. ETFs)

By shorting SPY, we reduce the beta exposure of the portfolio (and isolate the edge/idiosyncratic return/ allowing for an increase in absolute return rather than relative return)

Time-series-based models are more popular in the academic literature than in the real world, both because of interpretability and performance concern. Time-series betas are however used as additional characteristics in fundamental models.

Perhaps the easiest way to think about a factor model is as a superposition of effects on the stocks. A wave breaking on the shore does not have a perfect, sinusoidal shape. It is made of a large wave, and then of a few smaller waves riding on it, and then many ripples on top of the smaller waves. These effects sum up. Similarly, stock returns are the effect of a large shock (the market), then a few

smaller ones (sectors, the larger style factors), then a few even smaller ones.

There are three types of factor models:

1. Fundamental;
2. Time series;
3. Statistical.

Fundamental models rely on characteristics of the individual stocks and are commonly used by fundamental investors. They describe two features of returns:

1. Their expected returns;
2. Their risk.

shorting high-beta names and going long low-beta names - yields positive returns. This strategy is often termed "betting against beta (BAB)"

It is often said that "in a crisis, all correlations go to one"

Some institutional investors, for example, are not allowed to short stocks and to apply leverage to their portfolio. F&P suggest that these investors will seek higher returns by holding stocks with higher sensitivity to the market; higher beta should after all imply higher returns. By doing so, they provide excess demand for these stocks, which increase in price. Eventually the intrinsic value of the stock dominates, and the subsequent returns are negative. In this interpretation, the beta factor returns will be positive when we are in a "risk-on" environment (Figure 5.3). Investors are bullish, their risk appetite grows, and the way this is expressed in a portfolio is by buying riskier stocks linked to the economic trends.